

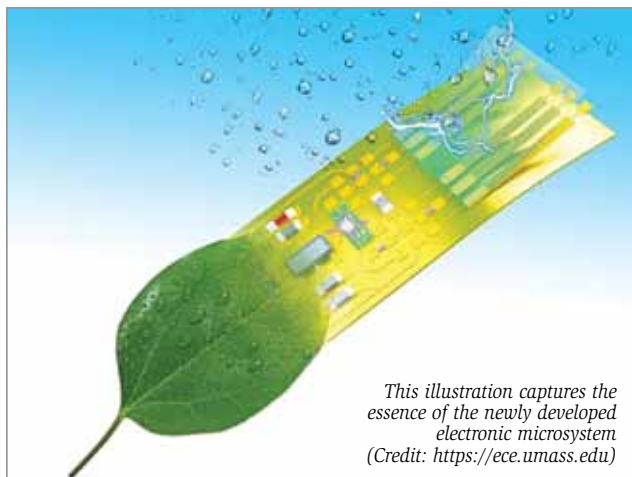
Google's AI tech that can accomplish chip design within a few hours

Scientists at Google Brain have introduced a deep reinforcement learning technique that allows the task of floor planning or optimal placement of different components of an electronic chip to be done within six hours, something that generally takes several months for a human to do. And the chip manages to excel in low power consumption and high performance. Although current software help to speed up the floor planning process, they become inefficient when the number of components in a chip grow and become complex. Google's engineers claim that this work would have major implications for the chip industry, enabling quick exploration of the possible architecture space for upcoming designs and easily customise chips for specific workloads.



An array of AI servers powered by Google TPU chips
(Credit: <https://venturebeat.com>)

Self-sustaining, intelligent electronic microsystems made using green tech

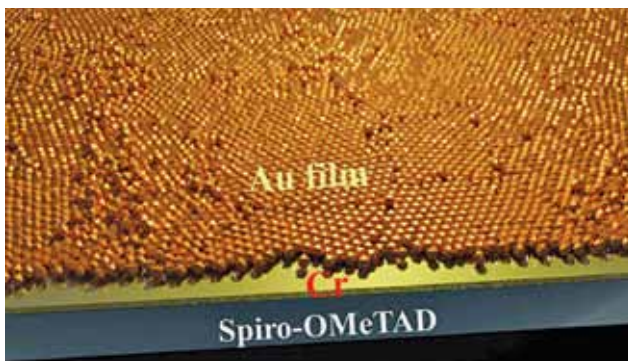


This illustration captures the essence of the newly developed electronic microsystem
(Credit: <https://ece.umass.edu>)

A research team from the University of Massachusetts Amherst has created an electronic microsystem that can intelligently respond to information inputs without any external energy input, much like a self-autonomous living organism. This is possible because the microsystem is constructed from a novel type of electronics that can process ultra-low electronic signals and incorporates a novel device that can generate electricity 'out of thin air' from the ambient environment. The key components are made from protein nanowires, a 'green' electronic material that is produced from microbes without generating e-waste. This research heralds the potential of green electronics to be made from sustainable biomaterials. Integrating protein nanowire based sensors, energy devices, and memristors of bio-amplitude functions yields flexible, self-powered, neuro-morphic interfaces that can intelligently interpret biologically relevant stimuli for smart responses.

Transparent electrodes that boost solar efficiency

Traditional solar cells are made from silicon but are fast approaching technological limits to create more efficient solar cells. Perovskite cells offer a promising alternative as stacking them on top of the traditional cells can create more efficient tandem devices. Therefore, researchers at Penn State University have developed new ultrathin metal electrodes that help to create highly efficient semitransparent perovskite solar cells. These can be coupled with traditional silicon cells to greatly boost the performance of both devices. When combined with a traditional silicon solar cell, the tandem device achieved 28.3% efficiency, up from 23.3% from the silicon cell alone. This basically means you are converting about 50 watts more sunlight for every square metre of solar cell material. The research represents a step toward developing completely transparent solar cells.



Scientists found using a chromium seed layer allowed them to grow ultrathin gold film that serves as a transparent electrode with good conductivity for perovskite solar cells (Credit: <https://news.psu.edu>)